PawMatch

By

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Problem definition + Stakeholders

According to BBC On average, the RSPCA received 374 reports of animal cruelty everyday in June to August 2024. The Royal Society for the Prevention of Cruelty to Animals (RSPCA) received 34,401 cruelty reports between June and August 2024, up from 25,887 in the same period the year before (BBC, 2024). These statistics show us how the amount of mistreated animals is increasing each year. I want to help reduce the likelihood of an animal receiving mistreatment from its owner.

My client is a friend at St Dominics called Matilde, who has a part time job looking after animals when their owners are away. She looks after a range of different animals from dogs and cats to chickens. We have various conversations about her work where she has made me aware how common it is for animals to be in unsafe environments, where signs of neglect are seen from the owners.

I aim to make an adoption management system which is aimed at adoption agencies but made specifically for one adoption agency, which will allow the staff to manage the animals up for adoption and their personal details and will allow any potential rescuers to see if they are suitable to adopt an animal.

Adoption agency staff are a key stakeholder for this project. This application will have a private part using an admin login for staff and requires an interface to allow them to interact with this software. This will contain details of all the animals and it will allow the staff to delete, modify, edit and add different animals and I will aim to also include queries from possible rescuers. They will also be able to view compatibility scores generated by the system when evaluating potential adopters. This system should reduce the amount of manual administration required and allows for a consistent form of assessing potential applicants.

Another key stakeholder is potential adopters who are the primary users of this system. Their experience with animals may vary significantly, ranging from first time pet owners to experienced animal carers. They will be using the public side accessed by a user's own logins stored within a large database, with hashing in order to keep the passwords safe if the database is breached. My system will also require a simple and intuitive interface allowing them to browse available animals, view information about each animal and complete an adoption questionnaire. This system will gather user information which will be processed by a compatibility scoring algorithm which will rank the right animals according to their suitability.

Another possible stakeholder is the animals. Although animals do not directly interact with the system, they are an important stakeholder as they are affected by the outcome of adoption decisions. This system's key aim is to improve animal welfare by encouraging more informed adoption choices and reducing likelihood of animals being placed in unsuitable homes.

Due to my clients experience caring for a variety of animals, she is skilled and knowledgeable about the environments required for an animal to live happily and safely and has asked me to incorporate the compatibility matcher with questions and a set criteria to examine whether they are suitable. Examples of the set criteria may include: the amount of available living space, access to a garden, experience with animals, income and more. Her feedback will be used throughout development to ensure the system meets the needs of animal welfare professionals.

Although many adoption websites allow users to browse animals, they often rely on their own judgement whether or not they are fit to adopt. This may lead to adoptions where the owner does not fully understand the commitment required which may lead to neglect. In some cases, unsuitable adoptions may result in animals being returned to shelters, creating additional pressure on adoption agencies and causing unnecessary stress for animals involved. This application should provide guidance whether a particular animal is compatible with their lifestyle, living arrangements, financial situation or previous experience. Additionally, current adoption agencies may use manual assessments and application forms by staff to which this may be timeconsuming and influenced by subjective judgement. My system will differ as it provides a consistent method of evaluating applicants through a scoring algorithm.

Use of Computational Methods

The most suitable solution for a problem like this is to create a computer based adoption matching system. This is because the problem includes storing animal data, adopter information, comparing different factors and producing recommendations. A manual system may require staff to compare each adopter with each animal individually, which could take significantly longer and will require more effort. Using a manual way of storing data may also lead to inconsistencies, missing information and errors, whereas using a database allows data to be searched efficiently and stored consistently. A computational solution allows this problem to be solved more efficiently by using algorithms, data validation, automated scoring and ensuring consistent data handling

Thinking Abstractly

The process of abstraction is removing unnecessary details in order to simplify the approach of a problem, making it easier to understand and solve. Ignoring insignificant details will reduce the amount of unnecessary coding and make the system easier to maintain. Within my system, it will be used in order to focus specifically on matching an adopter with an animal.
For example, the system does not need to store every detail about an adopter's personal life. It will only need to collect information that affects animal suitability such as the living space, garden access, experience with animals. Similarly, with the animal profiles it will only store useful information. It won't include its nicknames, or minor habits or favourite toy as this is not as important for the matching system.
This will make the system easier to design and implement as irrelevant details are ignored. This will also make the system more efficient by focusing on the most important factors that affect suitability.

Thinking Ahead

During the development of this system I will need to think ahead in order to identify the requirements of my user and relate to the issues they might face or be facing.
I have to anticipate how users would like their information to be presented in order to attract them to adopting an animal, it must also be fully accessible and easy to use.
I will also need to plan how the data will be stored before implementation. For example, animal records and adopter records must be stored persistently so information is not lost when the program closes. I will also need to consider validation as users should not be able to leave

empty fields or enter invalid data. This will help ensure that the matching algorithm is using accurate and correct information.
I will also need to think about the structures and designs that can be reused on different pages before I start developing my project. I also have to consider the technologies I will use to develop my system, as there are many different technologies I can use with each their own pros and cons.
Another key part of my system to which I will need to think ahead is the compatibility scoring system. This system should not simply recommend animals based on experience or breed. Instead it should compare multiple factors such as living space, experience, time available, children and animal requirements.

Thinking Concurrently

Processing multiple tasks at once will reduce the processing time of my app, reduce waiting time for my user and increase the overall efficiency of my system. When the user completes the adoption questionnaire, the system may need to retrieve animal records, compare the adopter answers with requirements, calculate compatibility scores and display results.
It is also important within the login page, where the credentials that are entered need to be validated and sanitised to prevent any injection attacks, whilst finding user profile. After this stage is cleared, multiple elements will need to be loaded at the same time.
The system also needs to be able to handle different users carrying out tasks at the same time. For example, an adopter could be viewing suitable animal matches while an admin user updates or adds an animal profile. I will need to design the system so these processes are independent and the data is able to remain consistent.
Concurrency may not be a main focus of my project, but it will allow me to design a system which is responsive and efficient as more animal records and adopter applications are added.

Thinking Procedurally with Decomposition

Decomposition is the process of splitting a large process into small more manageable sections. This will make the system easier to design, implement, test and improve. Instead of trying to build the whole system at once, I can separate it into different components.
The main components which build up my system will include animal records with their different cards, the adopter questionnaire with the matching algorithm, compatibility score calculation, results page, validation and the admin control page. The animal records section will store details about each animal and the adopters questionnaire collects information about users home and lifestyle. The matching algorithm will compare adopter answers with each animal's requirements and calculate compatibility score.

I will also split the application into the front end and back end sections. The front end is what the user or the admin user will experience, which will have the login forms and present the different features of the application. The backend is where all the logic will be processed and will include a relational database for secure storage of user data.
By using decomposition on my system it will allow each section to be developed and tested separately. For example, I can first test whether the animal records are being stored and added correctly, and if all the information is shown in the correct format. I can then test whether the adopter questionnaire collects all the information required and validates the inputs. Then I can test whether the matching algorithm calculates scores correctly. This way of testing will make errors easier to identify and fix.
Procedural thinking will allow me to further separate each function of front and backend and allow me to design my application using modules of reusable and readable code. The different subprograms will be kept in their own areas so I can work on them without affecting other areas. I will use this method in both the front and back ends of this project.

Thinking Logically

My program needs to be designed to make it simple to use but also logical to follow. This is important as users are guided through the adoption process clearly, from logging in to receiving suitable animal recommendations.
My system should hopefully follow a sequential process, for the user to view:

1. User creates an account or logs in using username and password
2. Once logins, users directed to main page, with the animals
3. The adopter will then complete a questionnaire
4. A compatibility score is shown for animals, and ranked from suitable to least suitable and the user can browse.

For the admin to view:

1. Admin user will login in using their username and passwords
2. The user will be directed to the admin control page
3. The animal is able to add, edit or delete records similar to a CRUD interface.
4. The system will validate the animal record to make sure required fields are completed and data in correct format and if it is valid will be saved to the database.
5. The updated animal information is then available for adopters to view and matching algorithms to use.

The system will also use logical decision making. For example, if the username or password is incorrect, the user cannot access their account. If required questionnaires fields are left blank, the system should display an error message. In the compatibility scoring algorithm, if an animal cannot live with children and the adopter has children then the compatibility score should decrease. If an animal requires a garden and the adopter has one then the score should increase.

This also shows that my system will use sequencing, selection, validation, iteration and sorting. It is not simply just displaying the animal profiles but processing data logically to produce suitable recommendations.

Research - Approach and Objectives

I will take a structured approach to my research by using both primary and secondary research in order to understand the problem from different perspectives and ensure my final system meets user needs and expectations:

1. I will inspect existing adoption websites, analysing their design, structure, features and adoption process, while looking at features I may want to include or avoid.
2. I will conduct an interview with my client (Matilde) to get a better understanding of what she wants within this project, and reflect this in the design of my system.
3. I will carry out a stakeholder survey to find out more about their preferences. I will use google forms and this will help me see if my problem is important and worth solving, and will help me in developing parts of the system such as the questionnaire.
4. I will assess the different technologies that can be used to produce my application.
5. I will identify software and hardware requirements for my project to run effectively.

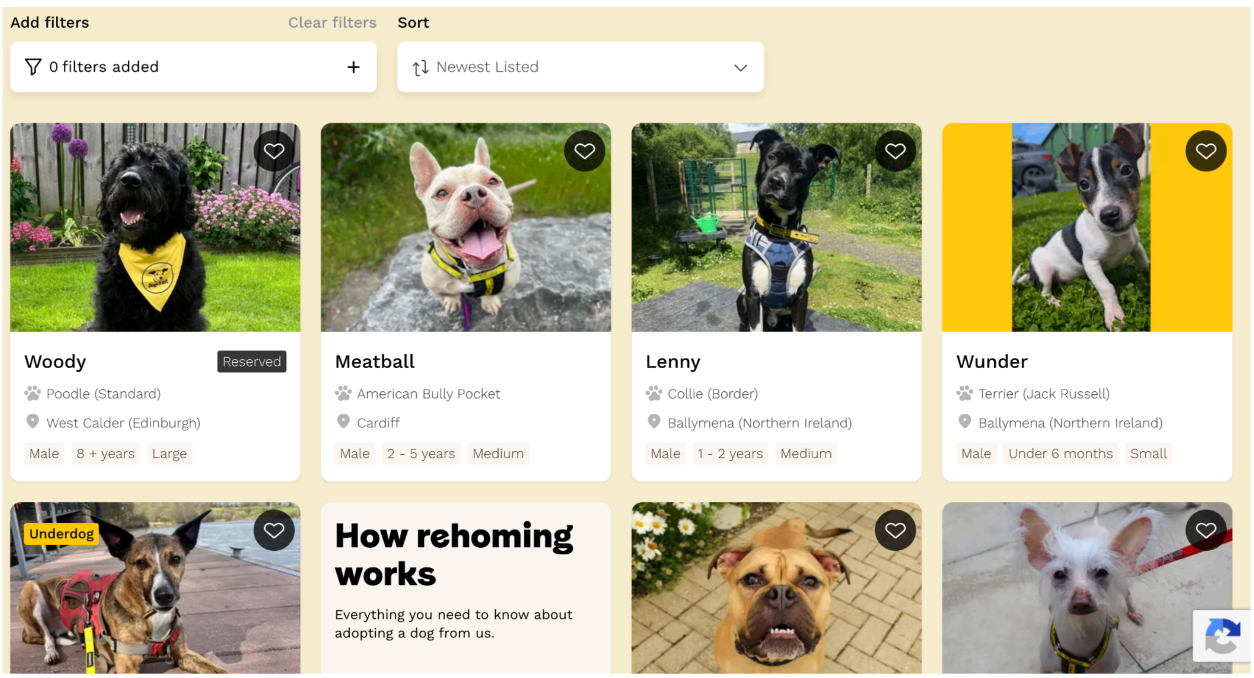
My research objectives are:

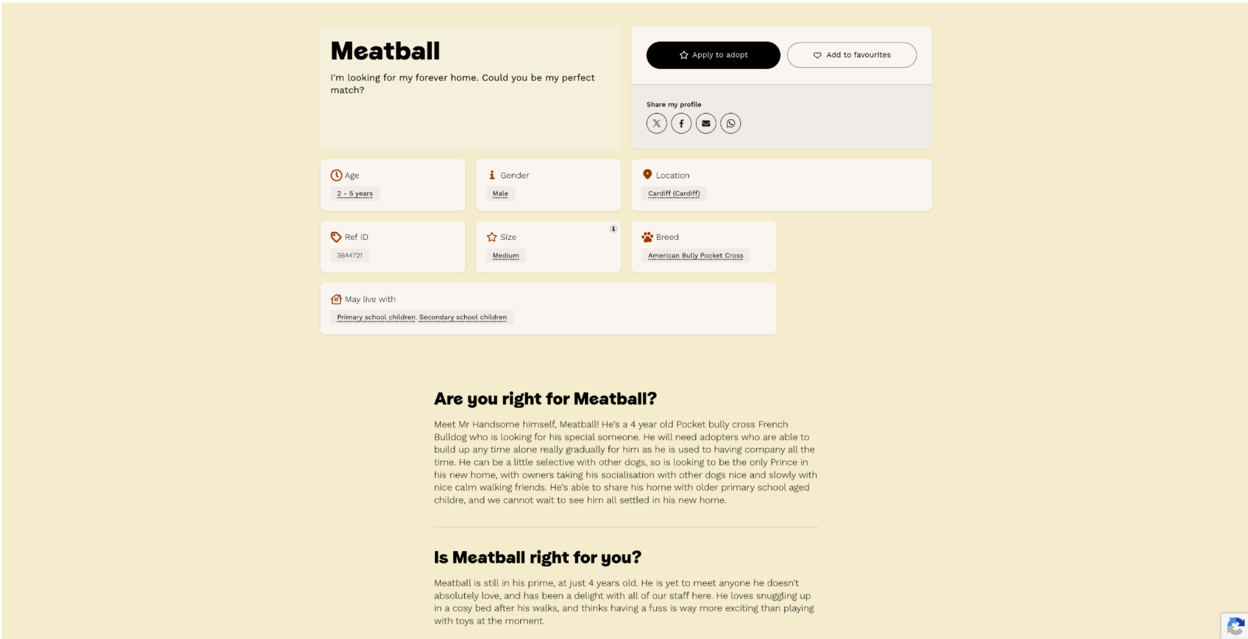
1. Understand current problems with adoption systems.
2. Identify which information needs to be stored about each animal
3. Identify which information needs to be collected from adopters
4. Find out what factors are most important when matching an adopter to an animal
5. Understand and take inspiration from existing adoption websites
6. Understand user expectations for the interface
7. Use the research findings to help decide the final features and requirements of my system

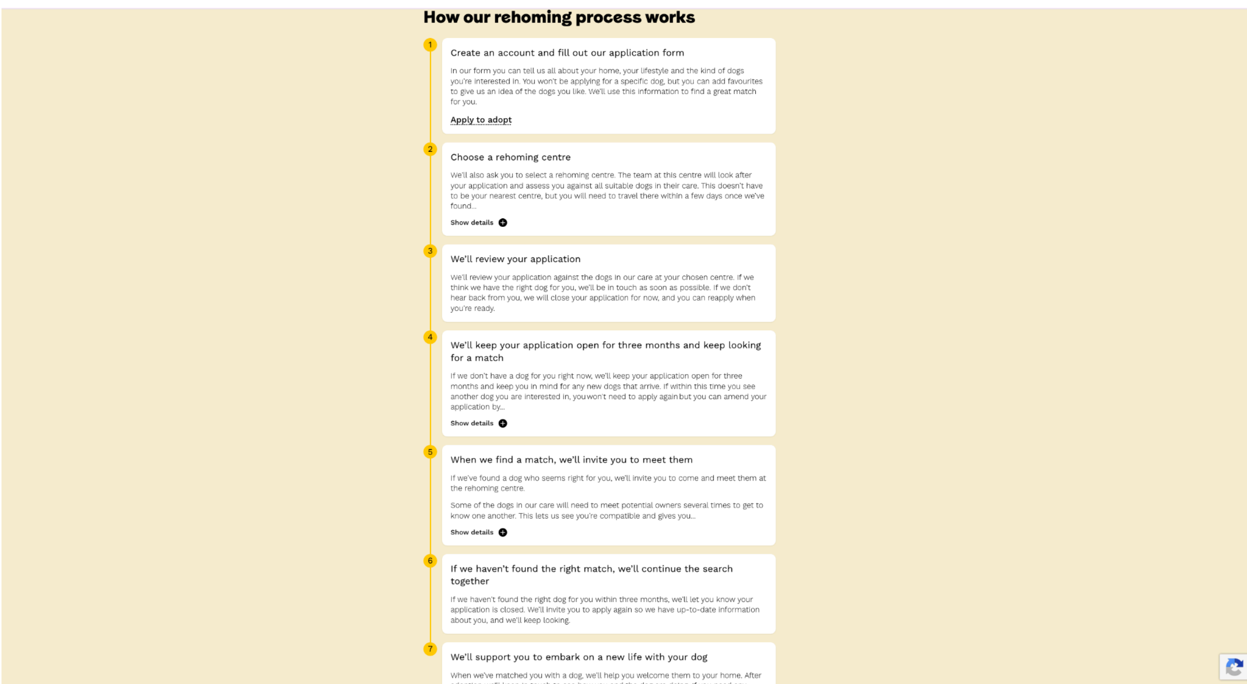
Research of Competitors

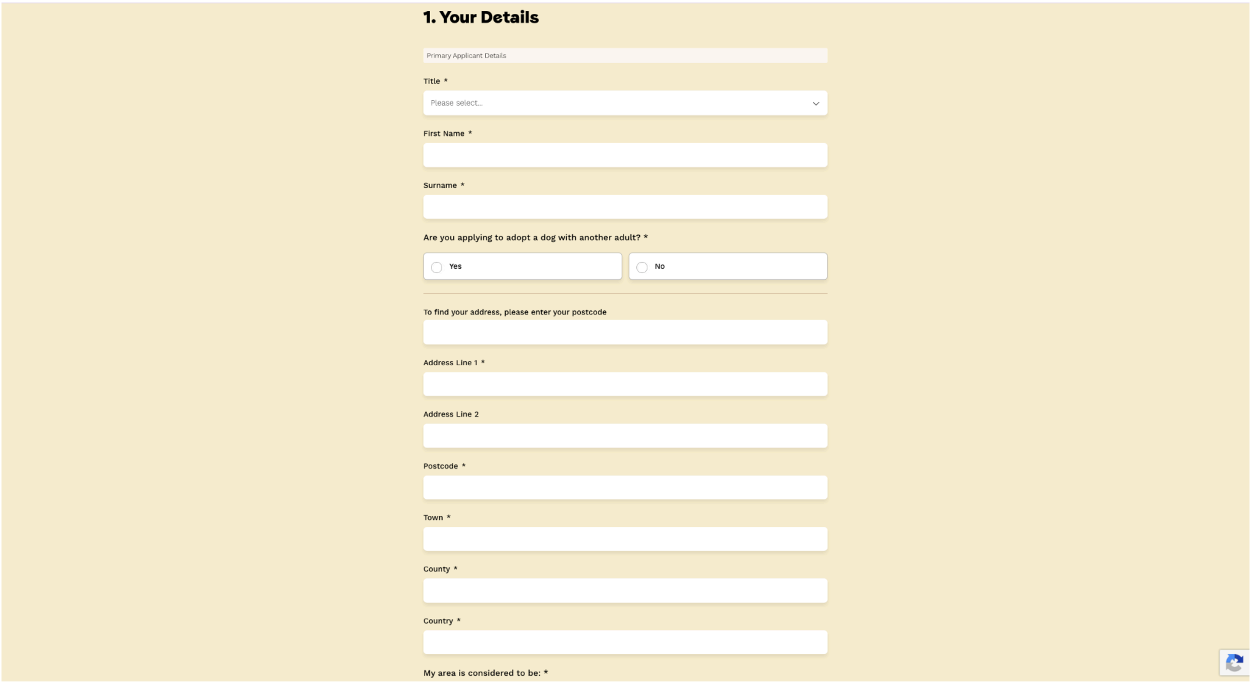
Example 1 - Dogs Trust

Dogs trust ‘offer practical solutions and support to help owners and dogs stay together wherever possible’ (Dogs Trust, 2026). They allow users to complete an application form which describes their home, lifestyle and preferences. Staff at the adoption home then compare applications with suitable dogs.




Meatball
I'm looking for my forever home. Could you be my perfect match?
Apply to adopt
Add to favourites
Share my profile
Age
2 - 5 years
Gender
Male
Location
Cardiff (Cardiff)
Ref ID
3644721
Size
Medium
Breed
American Bully Pocket Cross
May live with
Primary school children, Secondary school children
Are you right for Meatball?
Meet Mr Handsome himself, Meatball! He's a 4 year old Pocket bully cross French Bulldog who is looking for his special someone. He will need adopters who are able to build up any time alone really gradually for him as he is used to having company all the time. He can be a little selective with other dogs, so is looking to be the only Prince in his new home, with owners taking his socialisation with other dogs nice and slowly with nice calm walking friends. He's able to share his home with older primary school aged childre, and we cannot wait to see him all settled in his new home.
Is Meatball right for you?
Meatball is still in his prime, at just 4 years old. He is yet to meet anyone he doesn't absolutely love, and has been a delight with all of our staff here. He loves snuggling up in a cosy bed after his walks, and thinks having a fuss is way more exciting than playing with toys at the moment.


How our rehoming process works
1
Create an account and fill out our application form
In our form you can tell us all about your home, your lifestyle and the kind of dogs you're interested in. You won't be applying for a specific dog, but you can add favourites to give us an idea of the dogs you like. We'll use this information to find a great match for you.
Apply to adopt
2
Choose a rehoming centre
We'll also ask you to select a rehoming centre. The team at this centre will look after your application and assess you against all suitable dogs in their care. This doesn't have to be your nearest centre, but you will need to travel there within a few days once we've found...
Show details
3
We'll review your application
We'll review your application against the dogs in our care at your chosen centre. If we think we have the right dog for you, we'll be in touch as soon as possible. If we don't hear back from you, we will close your application for now, and you can reapply when you're ready.
4
We'll keep your application open for three months and keep looking for a match
If we don't have a dog for you right now, we'll keep your application open for three months and keep you in mind for any new dogs that arrive. If within this time you see another dog you are interested in, you won't need to apply again but you can amend your application by...
Show details
5
When we find a match, we'll invite you to meet them
If we've found a dog who seems right for you, we'll invite you to come and meet them at the rehoming centre.
Some of the dogs in our care will need to meet potential owners several times to get to know one another. This lets us see you're compatible and gives you...
Show details
6
If we haven't found the right match, we'll continue the search together
If we haven't found the right dog for you within three months, we'll let you know your application is closed. We'll invite you to apply again so we have up-to-date information about you, and we'll keep looking.
7
We'll support you to embark on a new life with your dog
When we've matched you with a dog, we'll help you welcome them to your home. After



Positives	Negatives
Dog profiles display clear information about the dog such as the age, gender, breed, size, location and whether the dog can live with children.	The system only supports dogs so people interested in different animals must use another service.
The profile also includes the dogs personality, behaviour and required home environment. This helps the user with making a decision on how the dog acts rather than from looks.	Users are not immediately told how well they match. Especially as suitability is assessed manually, so users have to wait.
Users only have to complete the application once rather than for each dog	The system does not explain which individual requirements the adopter meets or fails to meet
Users can add up to four favourite dogs.	The user must select a particular rehoming centre and be able to travel there when a match is found
Applications remain active for three months, so the user can be considered for dogs that arrive later.	Not every available dog is shown on the website meaning users cannot browse the complete set of possible matches themselves

Dogs are assessed by behavioral and veterinary staff before being matched, making the final decision more credible.	The bright yellow colour may be distracting and make it harder to read.
There is a strong yellow, black and white colour scheme making their branding recognisable.	The large photo graphs and sections mean that only a small amount of information is visible at once, so users must scroll frequently.
Large photographs are used for each dog to make it more appealing to adopt	Some pages are very long and contain several sections such as the 'About us' page, which may feel overwhelming.
There is plenty of white space between sections, preventing pages from appearing cluttered	The similar appearance of every dog card may make results look repetitive.
Important buttons such as apply to adopt and add to favourites stand out.	On smaller screens, the large images and vertically arranged content could require significantly more scrolling.

IMPACT ON MY SOLUTION

Dogs trust has influenced me to collect information about the adopters home, lifestyle and preferences only once and reuse it when comparing with different animals. I will also create similar animal cards which have detailed information about each animal. It will also contain another section with personality, behavior and home requirements.

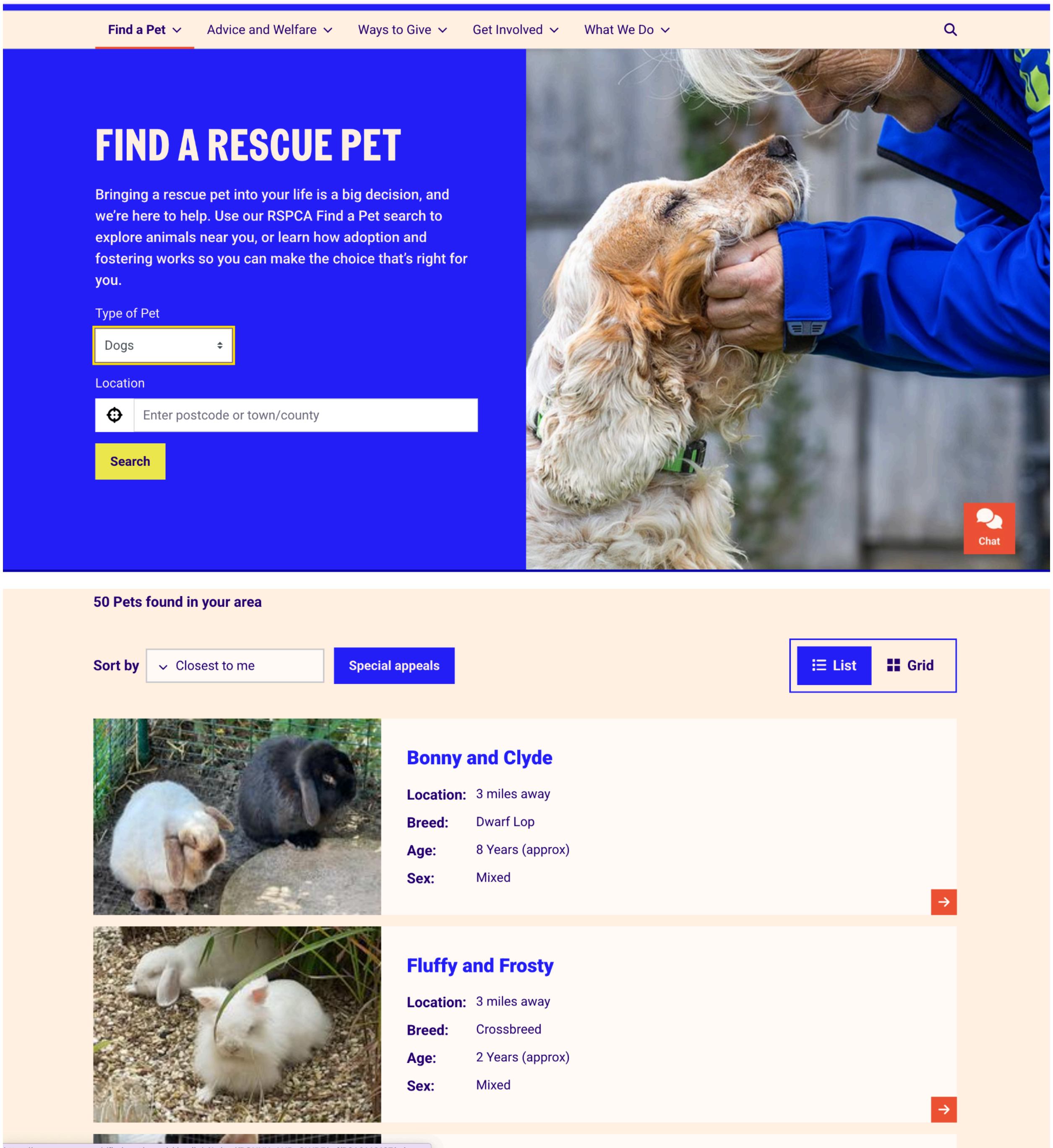
However, my system will differ as it allows for many different species and will automatically give partial feedback whether they are suitable, but ultimately leave the final decision to staff.

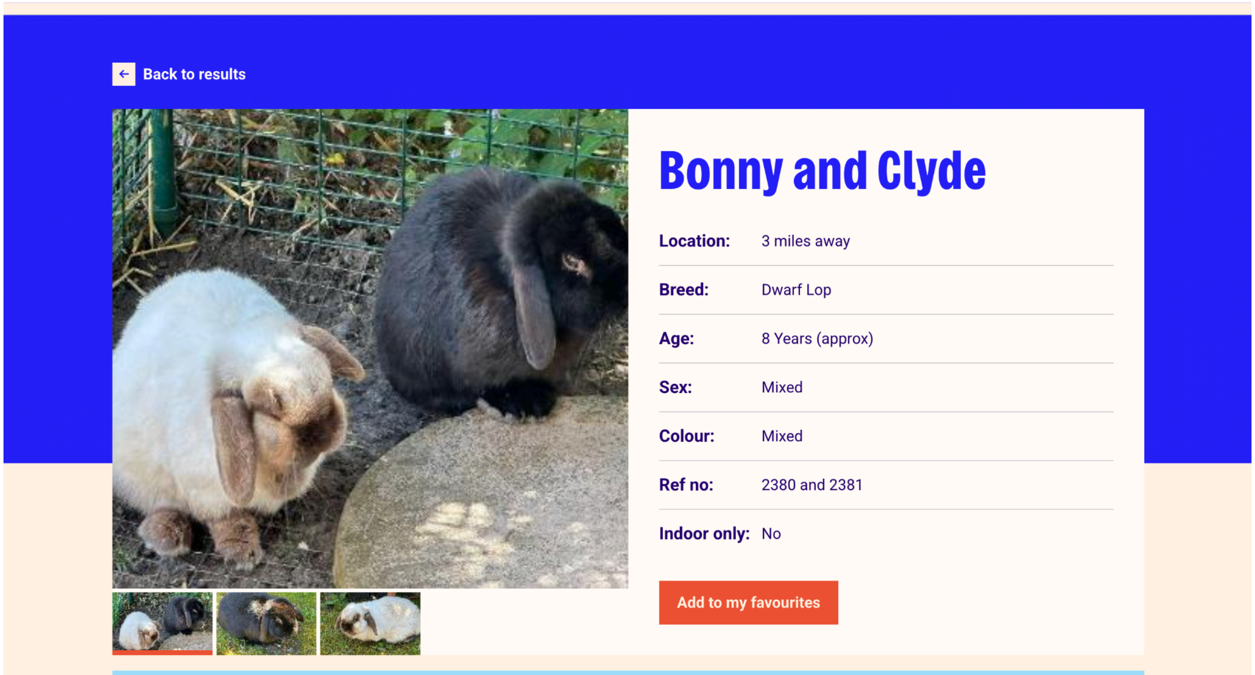
They have also influenced me to use a consistent colour scheme, clear headings with organised animal cards. I will make buttons easy to identify.

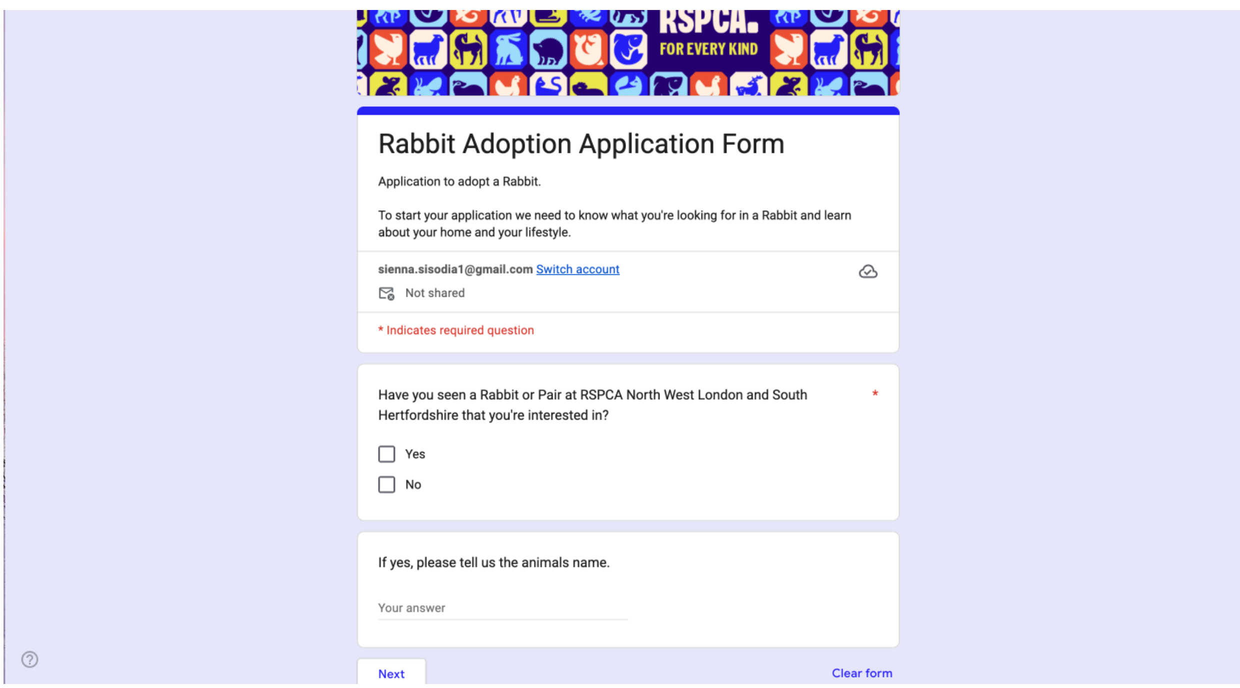
I will avoid bright colours, long pages and make mine less overwhelming and simple to use.

Example 2 - RSPCA

The RPSCA Find a pet website allows a user to search for several different types of animals. Its pages use photographs, clear heading and separate content sections to present animal information.







Positives	Negatives
Large animal photographs make the website visually appealing and immediately attract users' attention.	Applications are all reviewed manually by staff. The user does not receive an immediate result and should normally wait up to 48hours for a response. (Although this may

	vary between branches)
The colour scheme fonts and button styles remain consistent which allows the website to appear professional	The website contains many navigation options as it covers more than adoption, which may overwhelm some users. The number of filters and controls may make the search area appear crowded.
They support a wide range of species, rabbits, hamsters, guinea pigs. This is more similar to what I want within my system as it doesn't just support cats and dogs.	Some sections which are important and not important have a similar appearance which may make it hard to show what is important and allow it to stand out.
Each animal has an individual profile explaining needs, allowing adopters to consider suitability before applying. Applications also are reviewed by trained staff meaning factors for a computer to assess still can be considered.	Some important details are only available after opening the full animal profile.
They also have a feature for support after adoption, including behaviour services and pet care advice. This also could reduce chances of adoption failing after the animal entered a new home.	It may be hard to objectively analyse if they are suitable for the animal due to the lack of numeracy.
RSPCA may suggest alternative animals when the selected animal is unsuitable.	Procedures and costs may differ between RSPCA centres and branches, which does not allow for consistency which may confuse the user.
Search controls and filters are clearly labelled helping users understand what information they need to enter.	The published process requires applications to be started from an animal's profile. This may cause users to repeat parts of the process when interested in another animal.
Information is divided into sections with headings, helping users distinguish between different types of context.	Some centres only allow people to view animals by appointment after choosing a pet, this reduces the ability to visit freely and compare animals in person before applying.

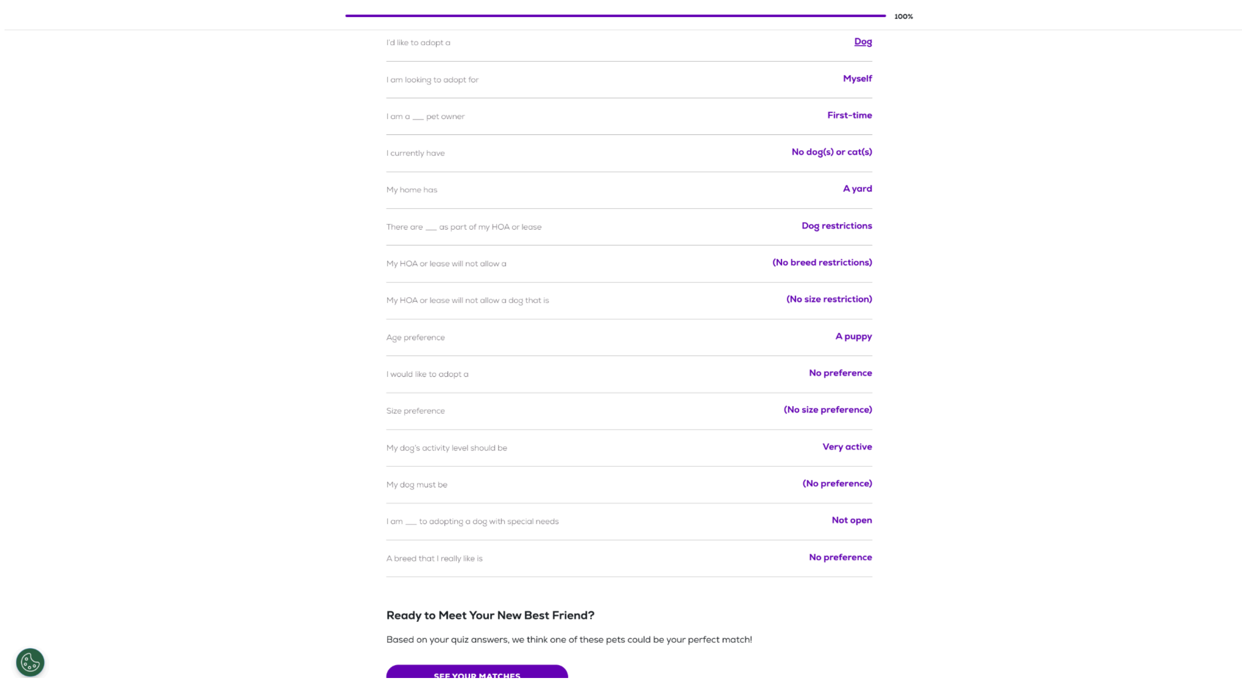
IMPACT ON MY SOLUTION

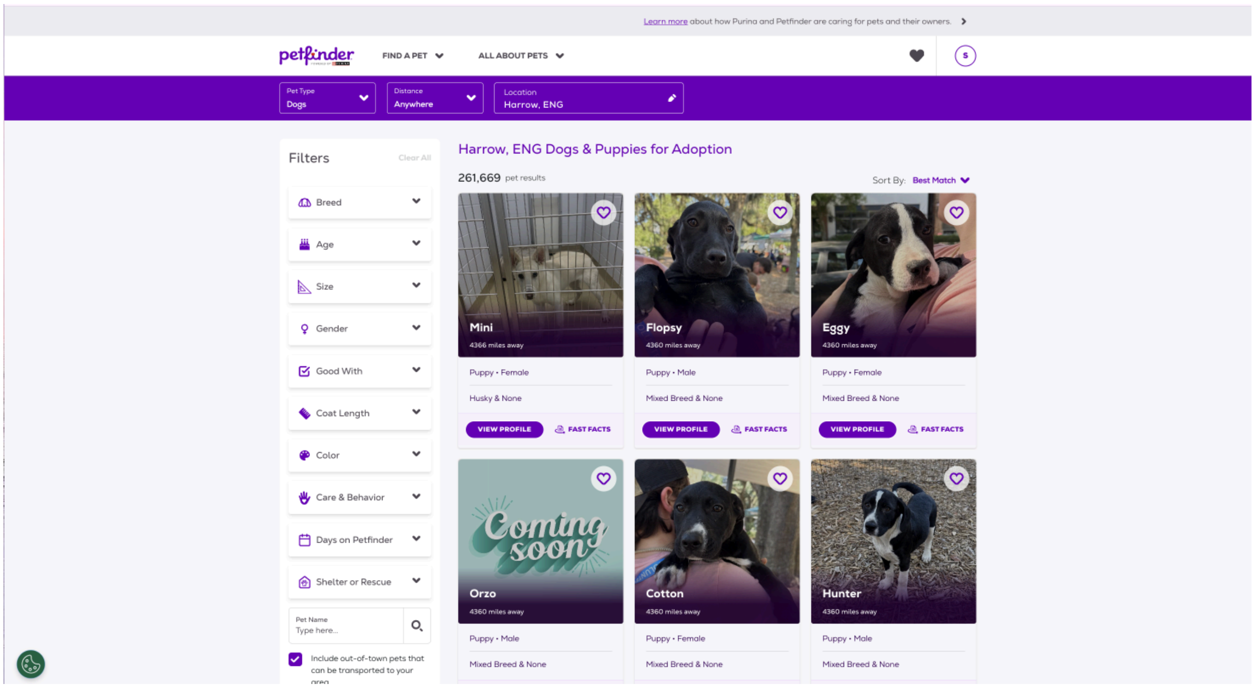
The RSPCA has influenced me to support a wide range of animal species and organise them into clear categories.I may include simple species categories, but the main method of finding suitable animals will be the compatibility questionnaire and ranked recommendations. It has also inspired me with its layout and visual design. I like its consistent color scheme with clear headings.

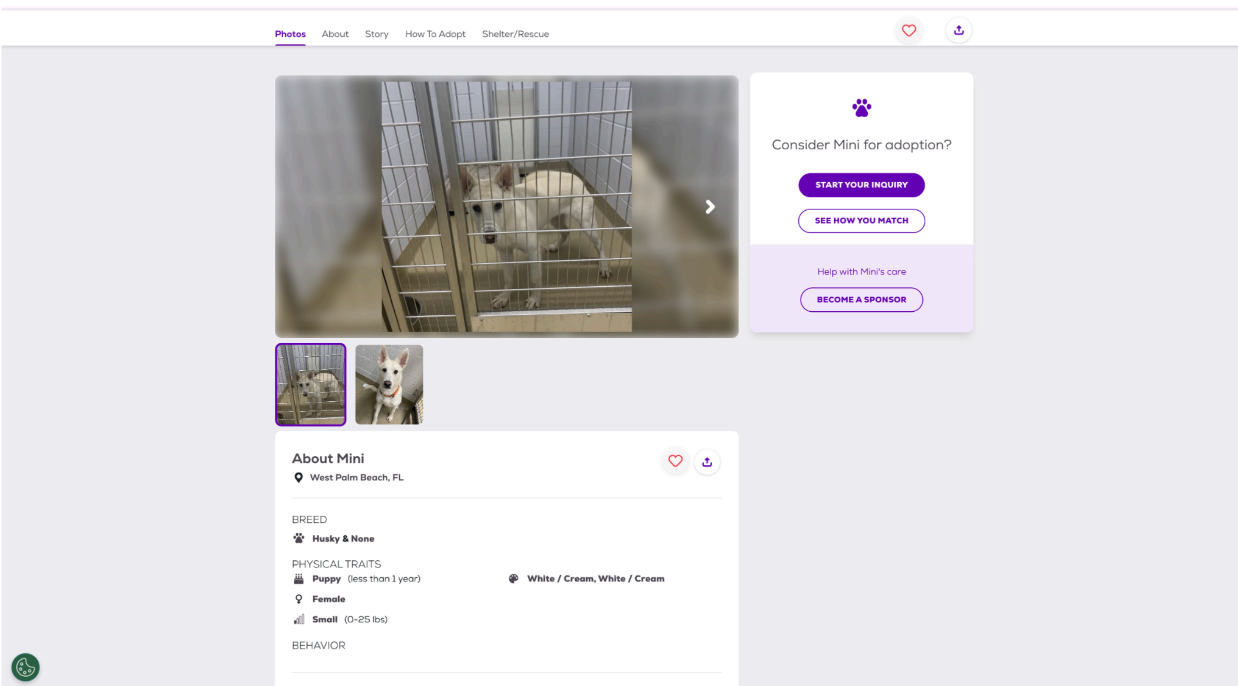
However, unlike the RSPCA, my system will calculate a numerical suitability score, explaining which requirements have and have not been met. It will also use a simpler navigation system focused solely on adoption, preventing users from becoming overwhelmed.

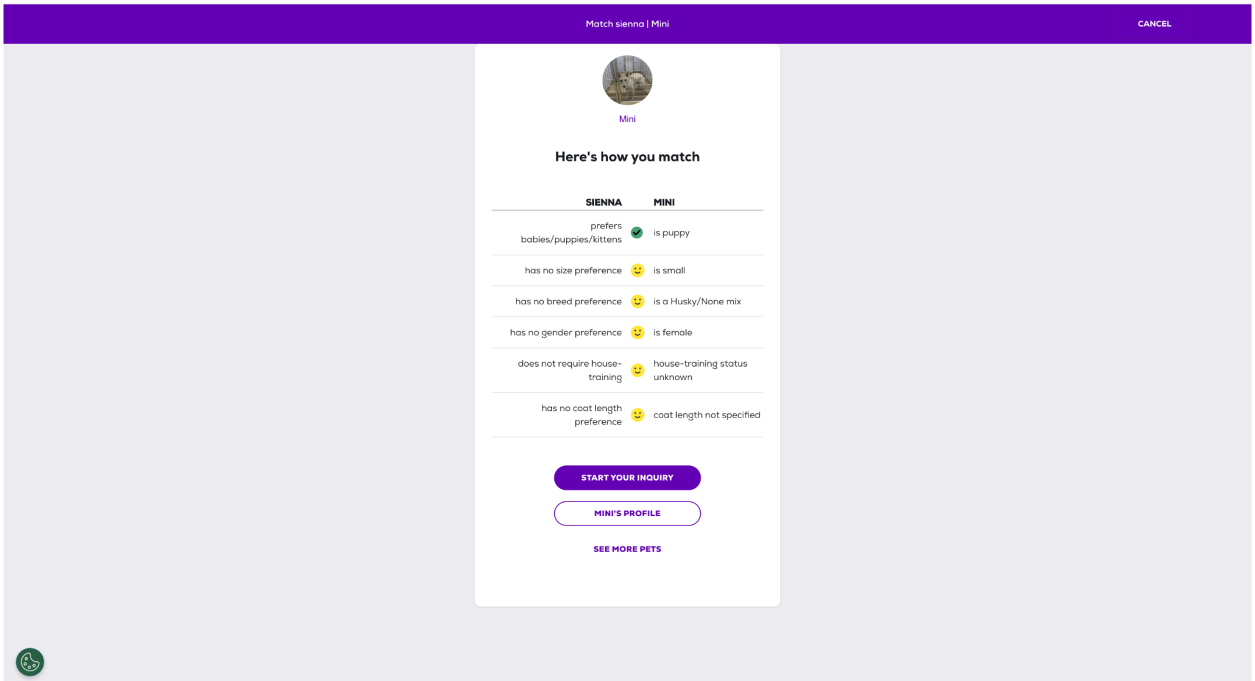
Example 3 - PetFinder

Pet finder is my closest competitor to my proposed system as it provides a short adopter questionnaire, and the system displays compatible pets. This website also includes a “See how you match” option. Animal listings contain information of the dogs such as age, housetrained ability and whether it can live with other animals. Users can also favourite animals.




Learn more about how Purina and Petfinder are caring for pets and their owners.
petfinder
FIND A PET
ALL ABOUT PETS
Pet Type Dogs
Distance Anywhere
Location Harrow, ENG
Filters
Clear All
Breed
Age
Size
Gender
Good With
Coat Length
Color
Care & Behavior
Days on Petfinder
Shelter or Rescue
Pet Name Type here...
Include out-of-town pets that can be transported to your area
Harrow, ENG Dogs & Puppies for Adoption
261,669 pet results
Sort By: Best Match
Mini
Flopsy
Eggy
Puppy • Female
Puppy • Male
Husky & None
Mixed Breed & None
VIEW PROFILE
FAST FACTS
Coming soon
Orzo
Cotton
Hunter


Photos
About
Story
How To Adopt
Shelter/Rescue
Consider Mini for adoption?
START YOUR INQUIRY
SEE HOW YOU MATCH
Help with Mini's care
BECOME A SPONSOR
About Mini
West Palm Beach, FL
BREED
Husky & None
PHYSICAL TRAITS
Puppy (less than 1 year)
White / Cream, White / Cream
Female
Small (0-25 lbs)
BEHAVIOR



Positives	Negatives
The search filters allow users to quickly narrow down the large number of available animals	The large number of filters and listings could overwhelm some users.
Animal cards are very clear and display a photo and important information such as the name, age and breed	It may not be as colourful and bright as the other examples.
The animal profiles are detailed and show information about the personality, medical needs and compatibility with children or other pets.	The automated matching appears to focus mainly on cats and dogs rather than equal support for every species.
Users can save animals as favourites so they can return to them later.	A visible suitability percentage is not provided making it harder for users to compare matches
The adopter questionnaire can provide more personalised recommendations then a basic search.	Information can be incomplete or marked as ‘unknown’ as each shelter is responsible for entering its own animal data
Important actions such as ‘start your enquiry’	The rescue organisation may ask the user to

and 'see how you match' are clearly labelled and easy to identify, allowing ease of use for the user.	complete another official application, causing information to be entered more than once. This also may mean the information provided may be inconsistent as different rescue organisations manage their own listings.
Pet finder brings listings from thousands of different shelters and rescues into one website.	Several actions appear on the same profile, including enquiring, matching and sponsoring which may compete for users attention.
Data can be saved by creating a login.	Some less important promotional content may distract users from main adoption features.
Bold headings divide profiles into clear sections such as behavior, health compatibility and adoption information.	As individual shelters manage their own listings, an animal's availability may not be updated immediately. Users might need to contact the shelter to confirm whether an animal is still available which may be an inconvenience for the user.

IMPACT ON MY SOLUTION

Petfinders system has strongly influenced the design of my system. I am inspired how the survey is divided into clear manageable questions rather than displaying a long form at once. I will use a similar step by step structure with clearly labelled answer options and progress indicator so the user knows how much of the questionnaire remains. I also like how it compares the adopters' preferences with animal characteristics in a clear table with visual symbols to show if each factor is compatible.

However, my system will provide more detailed feedback by displaying a numerical compatibility score and explaining which animal requirements have and have not been met. This will allow the several animals to be compared more objectively.

Overall Conclusion

My research shows that each competitor contains useful features, but none fully have all the requirements I want for my own system. Considering the design of dogs' trust, it has a recognisable and consistent interface where the information is presented very clearly. Dogs trust has a thorough application process even though it is only for dogs and does not provide immediate feedback. However, the frequent bright yellow may be visually distracting and long pages which can require excessive scrolling.

The RSPCA supports several species and organises animals into clear categories, but users must browse and select an animal before suitability is assessed. Their professional colour scheme makes it very recognisable and trustworthy, which may make people more intrigued to adopt. However because the website contains services beyond adoption, its navigation can appear crowded and some important information may not stand out. Users will not be given instant or numerical feedback as they browse for an animal then begin the application. Only after staff are able to manually assess the suitability.

Petfinder is the closest competitor to my proposed system as it uses an adopter questionnaire and compares the users preferences with individual animals. However, it does not provide a clear numerical score making it harder to compare with many animals. Information may also be inconsistent because individual rescue organisations maintain their own animal records.

Client Interview

1. **What factors do people fail to consider before adopting an animal?**

"I think a lot of people choose animals based on how cute they are without thinking how much work they can be. They often forget to consider things like how much space they have, how much time they can spend with their animal, how much it costs to look after, and whether they have experience caring for that type of animal. Some people also do not think about whether their home is suitable for the animals needs."

2. **What information should be stored about each animal?**

"Each animal should have basic information such as its name, age, breed, species, sex and a photo of the animal. It can also contain an extra information section such as personality, medical conditions, exercise needs, whether it gets along with children or other pets and special requirements other adopters should know about."

3. **How should the system respond if the user is not suitable for the particular animal?**

" I don't think the system should completely stop someone from adopting an animal, but it should warn them they might not be a good match. It should explain which requirements they don't meet and recommend other animals. This helps people make better decisions and reduce chances of animals ending up in the wrong home, but also not diverting people from adopting."

4. **Do you think users should be able to apply for adoption directly through the system?**

"No, I don't think users should be able to apply for adoption directly through the system. Although the system can help identify suitable animals, I still think adoption agencies should still carry out further checks in person. If staff are able to meet the adopter face-to-face they can assess personality, commitment and understanding of responsibility in owning a pet. It should aid the original process not completely take over that decision"

5. **What key tasks should the staff have on their side of the system?**

"The key features the staff should have is being able to add, edit and remove animal records, update animal information, upload photos and mark animals as available and unavailable for adoption by deleting. The database should also be kept up to date."

6. **What features would make the system easy to use for both staff and adopters?**

"I think the main thing is that it should be simple and not confusing. People should be able to find what they are looking for quickly without clicking through loads of pages. It should be easy to browse animals and use the questionnaire, and for staff it should be straightforward to update animal information. Clear buttons, a clean layout and simple instructions."

7. **What features should be displayed on the homepage?**

"I think the homepage should show information about how the adoption process works and a button to complete the questionnaire. The homepage should be welcoming and make it clear what the website is for."

8. **Should users be able to search for animals, if so what filters should be available?**

"Yes I think users should be able to search for animals as it makes it much easier to find a suitable pet. Filter categories could be the type of animal, breed, age, size and activity level, but mainly type of animal and size."

9. **What information should users provide when creating an account?**

"Basic information such as name, email, address, phone number and a password when creating an account. You could also include the age to ensure they are old enough to adopt the animal."

10. **Should some factors have a greater impact on the compatibility algorithm than others and if so which factors?**

"Yes I think some factors should matter more than others. Things like available space, allergies and whether there are young children in the house should have a bigger impact. For example, if someone is allergic to cats, they should not be strongly matched with a cat, even if the rest of their answers are suitable. Less serious factors such as preferred age or breed, should not be considered as important as the animals safety and wellbeing."

11. **Do you want the website to look a specific way? Any colours, fonts etc. It would also help if you gave specific examples of apps you like**

"I would like the website to have a cute and friendly look rather than it being too serious. Soft pink and pastel colours would make it feel more welcoming and enjoyable to use. I think the design should be simple but visually appealing, with lots of photos of the animals and clear buttons. The main thing is that it feels approachable and encourages people to engage with the adoption process."

12. **Do you have any security concerns regarding the app?**

“Yes, I think security is important because the system will store users' accounts and animal records. Passwords should be protected, and only staff should be able to add, edit or delete animal information. Regular users should only be able to browse animals and complete questionnaires, not change anything in the database.”

13. **Which devices do you want it to work on?**

“I want the system to work on both laptop and desktop computers, as these will be mostly used by adoption agency staff.”

14. **What should happen after a user receives their compatibility results?**

“I think the system should show users the results in a clear way and provide them with more information about any animals that are a good match. What happens after that would depend on how the adoption agency wants to use the system.”

15. **Are there any features that you believe are essential for the success of this system?**

“The compatibility scoring system is probably the most important feature as it helps match animals with suitable owners. Additionally, the animal profiles need to be detailed and up to date so the users will have all the information they need before making a decision”

Overall, the client interview provided valuable insight into the function and design requirements for this system. The client identified that many people fail to consider factors such as available space, time commitment, allergies and previous experience before adopting an animal. Therefore, these factors will be incorporated into the compatibility questionnaire and scoring algorithm. The compatibility algorithm should use weighted factors so that important requirements such as allergies, children, other pets and living space have a greater impact than softer preferences such as preferred breed or age. The client also highlighted the importance of maintaining and up to date detailed animal profiles and providing staff with ability to edit, add and remove animals through an administrative interface. For the design, the client wants a simple, easy to use interface with clear navigation, animal photographs and soft pastel colours to make it a friendly interface. The client also is very focused on security ensuring only staff can change details within the database regarding the animals.

Stakeholder Survey

This stakeholder survey received 27 responses, which gave me feedback from a range of potential users.

QUESTIONS:

1. **Have you ever owned a pet before?**

A. Yes
B. No

2. **If you did own a pet, did you adopt it from a centre?**

A. Yes
B. No
C. N/A

3. **How would you normally choose a pet?**

A. Appearance
B. Personality
C. Breed
D. Lifestyle Compatibility
E. Advice from experts

4. **How important do you think it is to research an animal before adopting?**

A. Very Important
B. Important
C. Neither
D. Not Important
E. Not very important

5. **Which factors do you think should be considered when matching an animal to an adopter? (Select many)**

A. Living Space
B. Garden Access
C. Income
D. Experience with animals
E. Time available
F. Presence of Children
G. Other pets

6. How important is it that an adoption website is easy to use?
   A. Very important (1)
   B. Slightly important (5)

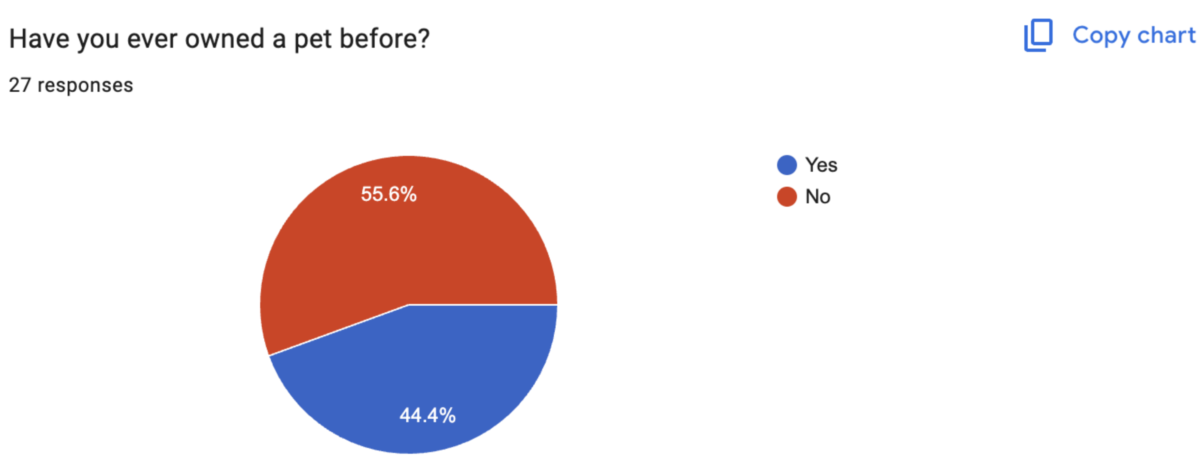
7. Would a compatibility score influence your decision when adopting an animal?
   A. Yes
   B. No

8. Do you think adoption agencies should use a compatibility scoring system?
   A. Yes
   B. No

9. How concerned are you about the welfare of adopted animals?
   A. Concerned (1)
   B. Not Concerned (5)

RESPONSES:

1.



Analysis:

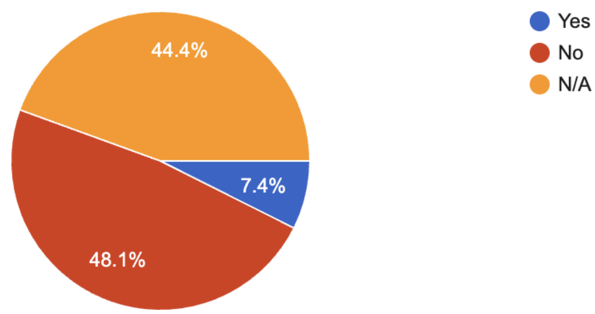
44.4% of respondents said yes, while 55.6% said no. This shows that my survey consists of a mixture of people with and without pet ownership experience. This is useful as my system needs to be able to support experienced and inexperienced adopters. This means the system should not assume that users will already understand animal care or adoption requirements.

2.

If you did own a pet, did you adopt it from a centre?

Copy chart

27 responses



Analysis:

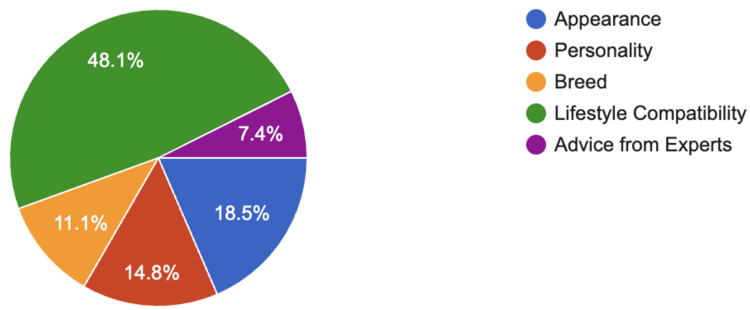
These results show that very few respondents have experience adopting from an adoption centre. This means my solution should clearly explain the adoption process and avoid making it confusing for users who have no experience.

3.

How would you normally choose a pet?

Copy chart

27 responses



Analysis:

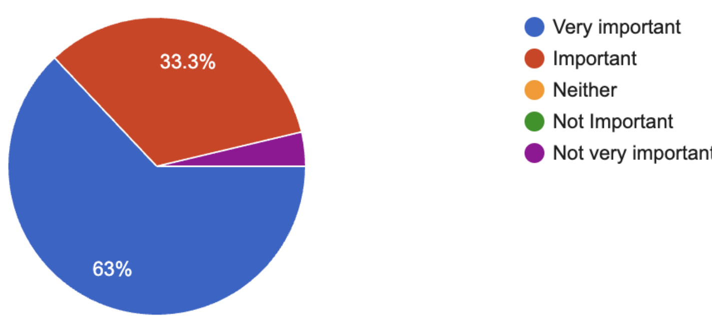
The most common answer was lifestyle compatibility, chosen by 48.1% of respondents. This shows most users already do care about which animal fits their lifestyle rather than just how the animal looks, even though a few care about the aesthetics of the animal. This will allow my system to be suitable for those looking to adopt.

4.

How important do you think it is to research an animal before adopting?

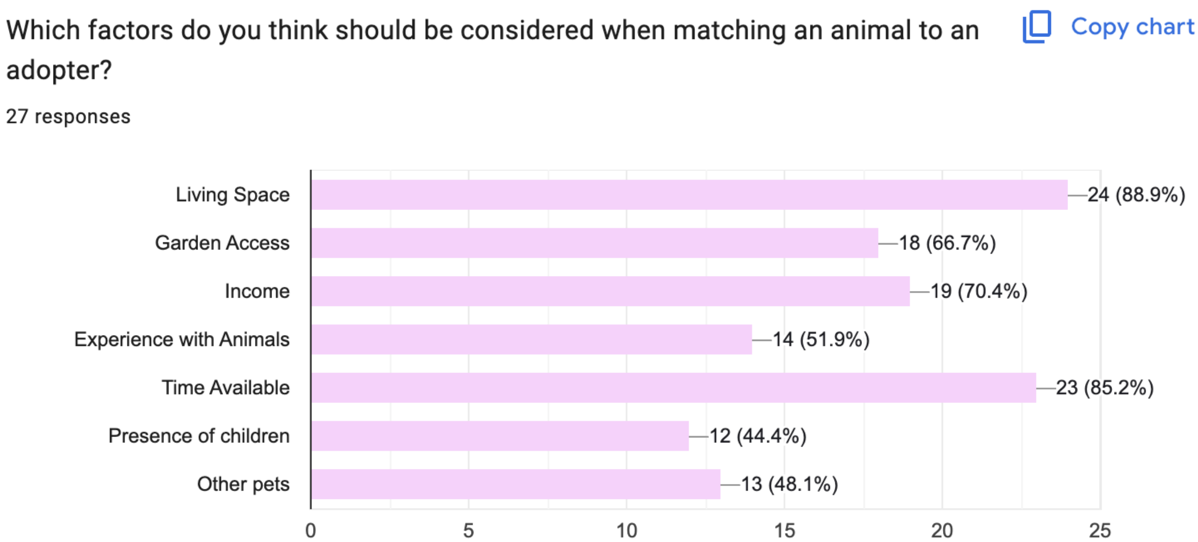
Copy chart

27 responses



Analysis:
63% of respondents said it was very important, and 33.3% said it was important. This means that 96.3% of respondents agreed that research is important before adopting an animal. This supports my idea that websites should include detailed animal information.

5.



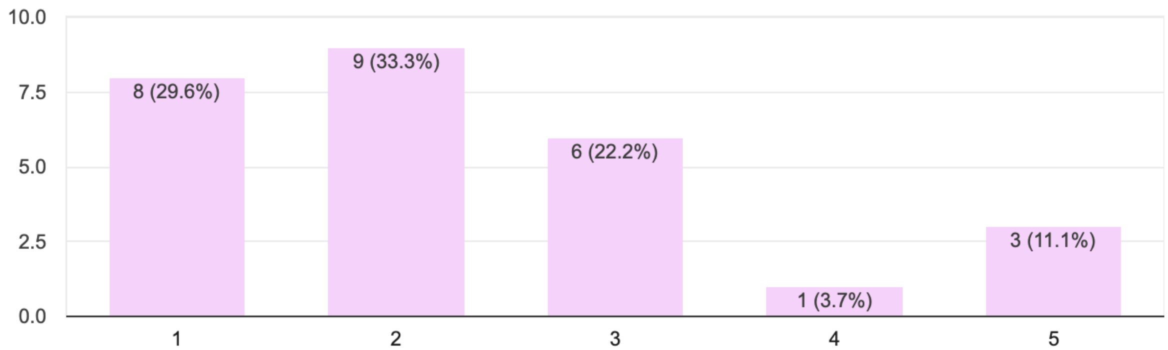
Analysis:
The most selected factors were living space at 88.9% and time available at 85.2% showing that respondents think an adopters environment and lifestyle are the most important parts of a successful match. Income and garden access were also highly chosen showing that practical care needs should be considered primarily.

6.

How important is it that an adoption website is easy to use?

Copy chart

27 responses



Analysis:

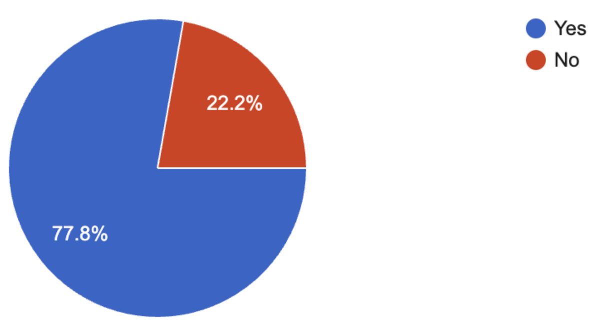
Most responses were grouped around the higher end of the scale, which suggests that ease of use is important to users. As a result, I would like my website to have a simple layout, clear navigation, readable text and easy process. I aim to make it as accessible as possible.

7.

Would a compatibility score influence your decision when adopting an animal?

Copy chart

27 responses



Analysis:

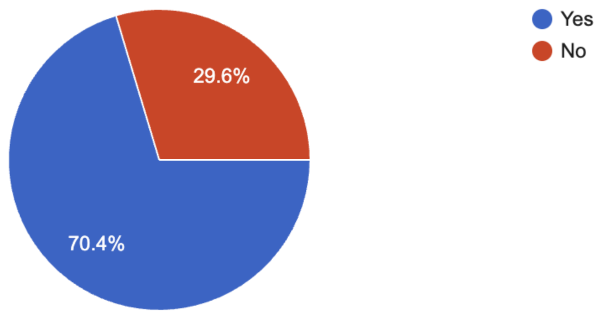
77.8% of respondents said yes and 22.2% said no. This shows that most users may find a compatibility score helpful when deciding whether an animal would be suitable for them.

8.

Do you think adoption agencies should use a compatibility scoring system?

Copy chart

27 responses



Analysis:

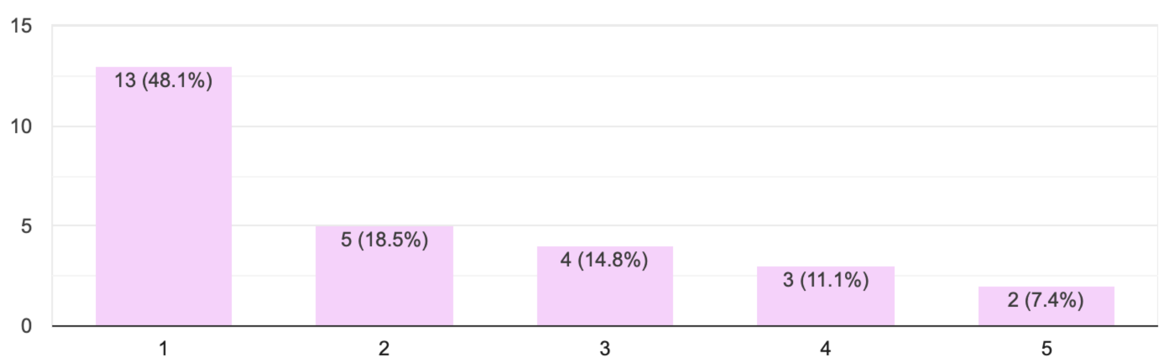
70.4% of respondents said yes, while 29.6% said no. This shows that respondents support the idea of adoption centres using this scoring system, but not everyone agrees. Therefore the compatibility score should be used as a guidance but not as the deciding factor.

9.

How concerned are you about the welfare of adopted animals?

Copy chart

27 responses



Analysis:

The largest group of respondents selected 1, with 48.1% and 18.5% selected 2. This shows that many respondents are concerned about animal welfare. This supports the purpose of this system as it aims to reduce unsuitable adoptions by helping users find animals and reduce chances of animals being placed in unsuitable homes.

Overall:
Overall, the survey shows that there is a clear need for a pet adoption system that helps users make informed decisions. Most respondents have shown that they agree that research is important before adopting an animal, and would be influenced by the compatibility score. This survey also shows that the important factors include living space, time available, income, garden access and experience with animals.

My solution should include detailed animal profiles, a user questionnaire with a compatibility scoring system, where lifestyle compatibility will be the most important and popular factor in choosing an animal. This system should also be easy to use as users will have different levels of experiences with pet ownership and adoption.

However, one limitation of this survey is that it only received 27 responses, so it may not represent all potential adopters, and some respondents may have limited experience with adoption centres. It still provided me with guidance about common user expectations, especially importance of ease of use, animal welfare and compatibility scoring.
This survey was insightful to help me understand what potential users would expect from an animal adoption system, especially the need for clear information, lifestyle based matching and a compatibility score.

Prefixes: E—Examples | Q—Client Interview|S—Stakeholder Survey

Features

#	Feature	Notes	Justification	Source
1	Clean user interface	Calm colour scheme, clear headings, identifiable buttons and a clear layout	Makes system easier to understand and use, especially for those not confident with technology	S6, E3, E1, E2
2	Navigation bar across different pages	Allows users to move between pages easily	Competitor websites used simple navigation, which helps reduce confusion and makes system more user friendly	E1, E2, E3, S6
3	User Home page	This system will include a clear home page introducing the purpose of adoption matching system	Simple starting point, and helps user understand system is focused on adoption suitability rather than browsing animals	E1, E2, E3
4	Animal preview Page	Users will be able to view available animals using compact animal cards. Each card will show key information such as name, species, breed, age and image.	Compact cards will make it easier for users to compare animals without needing to open every profile individually	E1, E2, E3
5	Animal Profile Page	Each animal will have a detailed profile page containing information such as name, species, breed, age, sex, personality, medical needs, exercise need and suitability requirements	The client stated that the animal profiles should show key details about the animal, which is also required for the matching algorithm	Q2, E1, E2
6	Multi Species Categories	Animals will be organised by species such as dogs, cats and rabbits	The RSPCA also has different animal categories, whereas dogs trust is just based around dogs.	E1, E2
7	Adopter	Users complete a questionnaire	The client stated that adopters	Q1, S4, S5

	Questionnaire	about their lifestyle and animal experience and will include many other answers which will contribute to the compatibility scoring algorithm.	often fail to consider important factors before adopting, this will allow them to reflect on their suitability before decisions are made.	
8	Step by step questionnaire layout	This questionnaire will be split question by question instead of being displayed as one long form, with dropdown options.	This makes the process easier to complete and reduces the chance of users feeling overwhelmed.	E3, S6
9	Input Validation	To check if data that is required is not left blank and the data is valid.	Preventing the incomplete or incorrect data being used by the matching algorithm, recommendations more reliable	Q15, S8
10	Compatibility scoring algorithm	The system will compare the adopter's answers with each animals requirements and calculate a numerical score.	The main computational feature, allows animals to be recommended on suitability	E3, S7, Q10
11	Ranked animal recommendations	After the questionnaire is submitted, animals will be displayed in order from most suitable to least suitable	Removes need for user to manually search through every animal	Q14
12	Requirements Breakdown	The system will show which requirements the adopters meet and which requirements they do not meet for each animal.	Helps users understand why an animal has received a particular score.	E3, Q3
13	Unsuitable match warning	If the user does not meet the requirements, the system will display a clear warning	The client said the system should not completely stop a user from adoption but warn them if they are unsuitable and why	Q3
14	User registration system	Users can create an account by registering with personal details such as name, email and password	Users can save progress, access their questionnaire and personalise their own experience	Q9
15	User and Admin Login and Logout system	Registered users will be able to securely login and logout their accounts and staff can login	User login ensures data is only accessible to the correct user and admin login prevents normal user	Q12

		securely to access management side of the system	from changing animal records and protects accuracy over database	
16	Admin animal management	Admins should be able to add, edit and delete animal records	Animal information should be kept up to date when new animals arrive, details change or animals are adopted	Q5
17	Database Storage	System will need to have a database alongside to store animal records, user accounts, admin login details and adopter questionnaire details	So data can be stored, searched, updated and used by the matching algorithm	Q15
18	Responsive streamlit design	I will use Streamlit layout tools so pages remain clear on different screen sizes.	This makes system easier to use on different laptops or tablets without information being cut off or difficult to read	Q6, E1, E2, E3, S6

Hardware and Software Requirements

The technology used to build this system is based on the features that my client requires. The list of requirements is as follows:

- Clean user interface
- User registration and login
- Admin login
- Database storage for animal records, user accounts and questionnaire answers
- CRUD functionality for admins
- Input validation
- Compatibility scoring algorithm
- Ranked animal recommendations

For this project, I will develop my system on my Macbook, which has the required hardware and software capabilities to develop, run and test my system.

Developer

Hardware:

Device: Macbook Air, 13 Inch, M3 2024
Processor: Apple M3 Chip
RAM: 8GB
Display: I will use the 13.6 inch built in retina display alongside a 13 inch monitor
Input: Keyboard and Trackpad
Internet Connection: This is required in order to install packages, access documentation and test deployment.
This hardware is suitable as my system will not require advanced graphics processing or a powerful server. The main processing will be form input, database queries, validation and the algorithm, which is perfectly suited for my laptop.

Software:

Operating system: MacOS Sequoia 15.0
IDE: Visual studio code
Programming Language: Python
User interface framework: Streamlit
Database: SQLLite
Database module: sqlite3

Browser: Chrome
I will use some additional libraries such as pandas which may be used to organise and display data in tables.

I have chosen python as my main programming language as it is the one i am most confident in, therefore I will not have to spend as much time trying to learn another language when python is suitable enough. It is suitable to create the matching algorithm, validating user inputs and all the functionalities my system requires. It allows me to be more efficient and focus on the computational parts of the system such as the comparison between adopter answers and animal requirements producing compatibility scores.

For my user and admin interface I will be using streamlit. It is simpler to use than using tkinter and it provides built in tools such as buttons, forms, dropdowns, columns, containers and page navigation. This allows me to create the pages without using a full HTML, CSS and JS website.

For data storage I will use SQLite as it is what I have the most experience in. It is suitable and easy to use to store data required for my system such as animal records, user accounts, admin login details, questionnaire answers and compatibility results.

User + Admins

My product will be a streamlit application, so users and admins will only require a device that accesses the system through any web browser., If deployed online, users will require an internet connection and they do not need to install any modules. They could access from a laptop, desktop computer or tablet, which may require a mouse, keyboard and a monitor to present the graphical user interface or even touchscreen if using a tablet.

Limitations

The system will be built in order for one specific animal shelter rather than a large organisation such as the RSPCA. This means that the database will only need to store the animals, users and admin accounts linked to that shelter. This is a large limitation as creating a system for multiple shelters would require extra features such as shelter admin accounts and location based searching and overall a larger database structure. This will be extremely difficult with the experience I have and would increase the complexity of my project. Furthermore, this would make development and testing less realistic within the time available.

The system will only be focusing on common adoption animals such as dogs, cats and rabbits rather than every possible species. This is due to different animals having different care requirements, and every possible animal would make the matching algorithm too broad and difficult to test within the time available. By focusing on a smaller number of species it will allow the compatibility algorithm to be more accurate and relevant. More species could be added within the future by expanding the animal requirements stored in the database.

Additionally the compatibility score is only there to provide guidance and will not replace the core decision making of adoption staff within the centre. Although it's able to compare adopters answers it will not be able to take in external factors such as personal circumstance or behavioural detail. It will also not be able to receive evidence to match their answers on the questionnaire. Therefore, the final adoption decision is made by staff, whilst the questionnaire is a guide in order to help those looking to adopt make a better decision.

The system will also not be able to complete the full adoption process as it cannot handle legal contracts, home checks, payments and final adoption approval. This would require more security, data protection and verification processes which widens the scope increasing development time. My system's main focus is helping users consider suitability before choosing an animal.

I cannot ensure the accuracy of the recommendations will be perfect every time, but rather they depend on the users answering the questionnaire honestly. If a user gives inaccurate answers on the questionnaire it will not reflect their true suitability. I will try to reduce this by using clear wording and validation but it cannot fully prevent users from entering inaccurate information.

The matching algorithm is rule based rather than using AI or ML. As a rule based algorithm is easier to explain, test and justify within the project. It will also be clearer to break down and show a user why an animal has or has not been recommended to the users. Additionally, machine learning would require a large dataset of previous adoptions which is not realistic for this project.

Animal records will need to be updated manually by admin users. This may lead to inconsistencies, or animals showing available when they are not. This is not something I can control but rather within the staff's responsibility to add edit and delete records so the database can be kept up to date.

Some advanced features such as animations, live chats would not be included and some of the formatting and design will be simplified. It would increase development and testing time without improving the main purpose of the system.

Prefixes: E—Examples | Q—Client Interview|S—Stakeholder Survey

Success Criteria

#	Feature	Requirements	Source
1	Navigation and Clean user interface	Pages must use a calm colour scheme, clear headings and identifiable buttons	**S6, E1, E2, E3**
		The interface must remain readable on laptop and tablet sized screens	**Q6, S6**
		The system must include a navigation menu to access home page, animal preview page, profile page, and login page	**E1, E2, E3, S6**
		The menu and interface style must remain consistent across all the pages	**E1, E2, E3, S6**
2	Home Page	Should explain the purpose of the animal adoption matching system, especially including that the system focuses on suitability rather than only appearance	**Q1, Q3, E1, E2, E3**
		Should have a clear button or navigation option that directs users to complete the adoption questionnaire	**Q1, S5, S6**
3	Animal preview page	The page should display every available animal using compact animal cards	**E1, E2, E3**
		Each card should show the species, name, breed, age and image	**Q2, E1, E2**
		The system must support at least 3 animal species including dogs, cat and rabbits	**E2**
		Users must be able to open a full animal profile from the preview page	**E1, E2, E3**
	Animal	Each profile must display the animals name, species, breed, age and sex	**E1, E2, Q2**

4	Profile Page	Each profile also must include personality, medical needs, exercise needs and suitability requirements	**Q2**
		Animal profile information must be retrieved from the database	**Q15**
		Users must be able to return to the animal preview page or results page	**S6, E1, E2, E3**
5	User and Admin registration and login page	Users must be able to create an account, using name, email and password and it can include other features such as age to ensure they are at a suitable age	**Q9, Q12**
		The system must reject registration if required fields are empty and the email format is invalid, by validating inputs	**Q12, Q15**
		Registered users must be able to login and logout using their correct account details	**Q12**
		The admin login details should allow admins to access their own section with the CRUD interface	**Q12, Q5**
		Incorrect Login details must display an error message and prevent access to the account	**Q12, S6**
6	Adopter Questionnaire Page	The questionnaire must collect information about the users home, lifestyle, garden access, children, other pets, time available and previous animal experience	**Q1, S5**
		The questionnaire must be split into clear sections or individual questions rather than appearing as one long form	**E3, S6**
		The questionnaire must use suitable input methods such as dropdowns, radio buttons or selection boxes but preferably dropdowns.	**E3, S6**
		The system must prevent the questionnaire from being submitted if important questions are unanswered	**Q15, S8**
		The system must compare adopters questionnaire with each animals requirements	**Q10, Q1, S5**
		The system must calculate a numerical compatibility score for each animal	**Q10, S7**

7	Results and recommendations page	The top 4 animals must be displayed from highest compatibility score to lowest at a minimum	**Q14, S7**
		The results page must show which key requirements the user meets and does not meet for each animal	**Q3, E3**
		The system must display a warning if the user does not meet important requirements for an animal	**Q3**
		The results page must explain that the compatibility scores are guidance and the adoption staff have the final say	**Q3**
8	Admin management page	This page should be accessed separately with a separate admin login	**Q12**
		Admin users must be able to add a new animal record with all the information required	**Q5, Q2**
		Admin users must be able to edit existing animal records and save the updated information to the database	**Q5, Q15**
		Admin users must be able to delete animal records when animals adopted or no longer available	**Q5**
		Normal users must not be able to access the admin management page	**Q12**
9	Database system	The database must store animal records, user account details, admin login details and questionnaire answers	**Q15**
		The database must allow animal records to be added, viewed, edited and deleted through the admin system	**Q5, Q15**
		Questionnaire answers must be stored or processed so that they can be used by the compatibility scoring algorithm	**Q10, Q15**
		Data must be retrieved correctly so animal profiles and recommendations display accurate information	**Q15, Q2, Q10**

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